

Connectivity layer: This layer ensures seamless communication between IoT devices and the middleware. Common protocols include MQTT, CoAP, HTTP, and Bluetooth.

Edge layer: This processes data closer to the source (IoT devices) to reduce latency and improve real-time decision-making.

Middleware mediates between device layers and IoT data consumers. APIs for interactions with the middleware are based on standard application protocols. API endpoints for accessing the data and/or services should be searchable via an open catalogue and should contain linked metadata about the resources.

Data processing performs data translation, aggregation, and data filtering on incoming data. It is responsible for real-time decision-making at the edge. The database needs to support high-speed reads and writes with sub-millisecond latency.

IoT data stream normalises the data to a common format and sends it to enterprise systems. Cloud services provide scalable storage, computing, and networking resources to support IoT applications. Data analytics uses advanced analytics and machine learning capabilities to derive insights from IoT data.

IoT middleware supports authentication of users, organisations, applications and devices. It assists functionalities like certificates, password credentials, API keys, tokens, etc. It also supports single sign-ons, time-based credentials, application authentication (via signatures), and device authentication (via certificates).

Middleware manages logging of system debugging and auditing details. It helps to track the status of the various services, APIs, etc, and administer them.

Key open source IoT middleware platforms

IoT middleware is categorised as:

- Application-centric

- Platform-centric
- Industry-specific

Here are some of the top open source middleware platforms for IoT based applications.

Kaa: This platform-centric middleware manages an unlimited number of connected devices with cross-device interoperability. It performs real-time device monitoring, remote device provisioning and configuration, and collection and analysis of sensor data.

This microservices-based portable and highly available IoT platform supports on-premises, public cloud and hybrid models of deployment.

Kaa is built on open components like Kafka, Cassandra, MongoDB, Redis, NATS, Spring, React, etc.

SiteWhere: This platform-centric middleware provides ingestion, storage, processing, and integration of device data. It supports multi-tenancy, MQTT, AMQP, Stomp, CoAP and WebSocket. Built on open source technology stacks like MongoDB, Eclipse Californium, InfluxDB, and HBAs, it seamlessly integrates with Android, iOS, and multiple SDKs.

IoT SyS: IoT SyS is platform-centric and industry-specific middleware that uses IPv6 for non-IP IoT devices/systems. It provides interoperable web technologies for sensors and actuator networks, and is used in smart city/smart grid projects.

DeviceHive: This cloud-agnostic, microservices-based platform-centric middleware is used for device connectivity and management, and can connect to any device via MQTT, REST or WebSocket.

It supports Big Data solutions like Elasticsearch, Apache Spark, Cassandra and Kafka for real-time and batch processing.

EclipseIoT (Kura): This platform provides device connectivity, data transformation, and business logic with intelligent routing, edge processing/ analytics and real-time decisions.

Zetta: This application-centric middleware is a service-oriented open source IoT platform built on Node.js. It can run on cloud platforms like Heroku to create geo-distributed networks and stream data into machine analytics like Splunk.

Node-RED: This is a visual programming tool that simplifies the creation of IoT applications.

MainFlux: This application-centric middleware provides solutions based on the cloud. Containerised by Docker and orchestrated with Kubernetes, its architecture is flexible and allows seamless integration with enterprise systems like ERP, BI and CRM. It can also integrate with databases, analytics, back-end systems and cloud services easily. It supports REST, MQTT, WebSocket and CoAP.

DSA: Distributed Services Architecture (DSA) facilitates decentralised device intercommunication, allowing protocol translation and data integration to and from third party data sources.

OpenRemote: This application-centric middleware is used to connect any device regardless of vendor or protocol to create meaningful connections by converting data points into smart applications.

OpenIoT: This is an application-centric open source middleware for pulling information from sensors. It acts as sensing-as-a-service for deploying and providing services in a cloud environment.

ThingsBoard: This is an open source IoT platform for data collection, processing, data visualisation and device management. It supports IoT protocols like MQTT, CoAP and HTTP with on-premises and cloud deployment. It is horizontally scalable and stores data in Cassandra DB, HSQLDB and PostgreSQL.

NATS: Developed as microservices architecture, NATS.io is a simple, secure, and high-performance open source messaging system for cloud native solutions.